(L)297mm

Front

(W)210mm

pharmaniaga®

Allopurinol

Tablet 300 mg

DESCRIPTION
A yellow, round, biconvex tablet with Pharmaniaga icon on one side and plain on the other.

COMPOSITION
Each tablet contains Allopurinol 300 mg.

ACTIONS
Allopurinol inhibits the action of the enzyme xanthine oxidase, thus reducing the oxidation of hypoxanthine to xanthine to uric acid. The urinary purine load, normally almost entirely uric acid, is thereby divided between hypoxanthine, xanthine and uric acid, each with its independent solubility.

This result in the reduction of uric acid concentrations in both plasma and urine, ideally to such an extent that deposits of uric acid are dissolved or prevented from forming. At low concentrations of allopurinol, which is an analogue of hypoxanthine, acts as a competitive inhibitor of xanthine oxidase, but at higher concentrations both allopurinol and its active metabolite oxypurinol acts as non-competitive inhibitors of the enzyme.

By lowering both serum and urine concentrations of uric acid below its solubility limits, allopurinol prevents or decreases urate deposition, thereby preventing the occurrence or progression of both gouty arthritis and urate nephropathy. In patients with chronic gout, allopurinol may prevent or decrease tophi formation and chronic joint changes, promote resolution of existing urate crystals and deposits, and after several months of therapy, reduce the frequency of acute gouty attack. Also reductions in urine urate concentration prevent or decrease the formation of uric acid or calcium oxalate calculi.

PHARMACOKINETICS
Allopurinol is absorbed from the gastro-intestinal tract after oral administration and is reported to have a plasma half-life of about 1 to 3 hours. It is converted primarily in the liver to oxypurinol (aloxanthine) which is also an inhibitor of xanthine oxidase with a reported half-life of 12 to 30 hours (average 15 to 18 hours) in patients with normal renal function, although this is prolonged by renal impairment.

Both Allopurinol and oxypurinol are conjugated to form their respective ribonucleosides. Allopurinol and oxypurinol are not bound to plasma proteins.

Excretion is mainly through the kidney, but is slow since oxypurinol undergoes glomerular filtration as well as tubular reabsorption. About 70% of a daily dose may be excreted in the urine as oxypurinol and up to 10% as allopurinol, prolonged administration may alter these proportions, due to allopurinol inhibiting its own metabolism. The remainder of the dose is excreted in the feaces. Allopurinol and oxypurinol have also been detected in milk.

INDICATIONS
Allopurinol is used to treat primary or secondary hyperuricaemia associated with chronic gout, uric acid nephropathy, recurrent uric acid stone formation, certain enzyme disorders, and neoplastic disease with high cell turnover rates in which high urate levels occur either spontaneously or after cytotoxic therapy.

CONTRAINDICATIONS
History of hypersensitivity to allopurinol or treatment of an acute attack of gout. Additionally, allopurinol therapy should not be initiated for any purpose during an acute attack.

ADVERSE REACTION
Following initiation of allopurinol for gouty arthritis, the most commonly encountered adverse effect is a temporary increase in the frequency of acute gout attacks. The occurrence of such reactions may be reduced by initiating therapy with a low dose that is gradually increased until desired effects is obtained and by administration of prophylactic doses of colchicine or a non-steroidal anti-inflammatory drug.

Rashes are generally masculopapular or pruritic, but more serious hypersensitivity may occur and include exfoliative rashes, the *Stevens-Johnson syndrome* (SJS), *toxic epidermal necrolysis* (TEN), and *drug reaction with eosinophilia and systemic-symptoms* (DRESS).

Allopurinol is often used in asymptomatic hyperuricemia. It is therefore, recommended that allopurinol be withdrawn immediately if a rash occurs.

Further symptoms of hypersensitivity include fever and chills, lymphadenopathy, leucopenia or leukocytosis, eosinophilia, arthralgia, and vasculitis leading to renal and hepatic damage and, very rarely, seizures.

Haematological effects include thrombocytopenia, aplastic anaemia, agranulocytosis and haemolytic anaemia. These hypersensitivity reactions may be severe, even fatal, and patients with hepatic or renal impairment are at special risk.

Hepatotoxicity and signs of altered liver function may be found in patients not exhibiting hypersensitivity.

Many other adverse effects have been noted rarely and include paraesthesia, peripheral neuropathy, alopecia, gynecomastia, hypertension, taste disturbances, nausea, vomiting, abdominal pain, diarrhoea, headache, malaise, drowsiness, vertigo, and visual disturbances.

Patients with gout may have an increase in acute attacks on beginning treatment with allopurinol, although attacks usually subside after several months.

Warnings
Prophylactic colchicine or NSAID (not aspirin or salicylates) should be administered until at least 1 month after hyperuricaemia is corrected.

Allopurinol should be discontinued at the first appearance of skin rash or other signs which may indicate an allergic reaction. Hypersensitivity to allopurinol usually appears after some weeks of therapy, and more rarely immediately after beginning treatment. In some instances, a skin rash may be followed by more severe reactions such as exfoliative, urticarial and purpuric lesion as well as Stevens-Johnson syndrome, and/or generalized vasculitis, irreversible hepatoxicity and even death.

A cautious re-introduction at a lower dose may be attempted when mild skin reactions have cleared. Allopurinol should not be introduced in those patients who have experienced other forms of hypersensitivity reaction.

Allopurinol should be administered with care to patients with renal or hepatic impairments, and dosage may need to be reduced.

In all patients receiving allopurinol it is advisable to ensure adequate fluid intake to maintain a urinary output of not of less than 2 litres a day and for the urine to be neutral or slightly alkaline.

In neoplastic conditions, treatment with allopurinol should be commenced before cytotoxic drugs are given.

> If you experience side effects such as rash, fever, sore throat, or irritation of the eyes, **stop taking this medication IMMEDIATELY** and consult your doctor/pharmacist.

PRECAUTIONS
Pregnancy
Although adequate and well-controlled studies in humans have not been done, three reports indicate no evidence of birth defects in off-spring of women receiving allopurinol during pregnancy. Should allopurinol be indicated during pregnancy, the risk to which the foetus may be exposed must be weighed against the risk of the maternal disease process.

Breast-feeding
Allopurinol and oxypurinol are distributed into breast milk. Whether this toxic medication may cause adverse effects in the nursing infant has not been determined. However, problems in humans have not been documented.

Paediatrics
Appropriate studies performed to date have not demonstrated paediatrics-specific problems that would limit the usefulness of allopurinol in children. However, use of allopurinol in paediatric patients has been limited to children with certain rare inborn errors of purine metabolism or hyperuricaemia secondary to a malignancy or cancer therapy.

Geriatrics
No information is available on the relationship of age to the effects at allopurinol in geriatrics patients. However elderly patients are more likely to have age related renal function impairment, which may require adjustment of the dose and/or dosing interval in patient receiving allopurinol.

DRUG INTERACTIONS

1. Anti-coagulants, coumarin-or indandione derivative:
   Allopurinol may inhibit enzymatic metabolism of the anti-coagulant leading to potentiation of the anti-coagulant effects; dosage adjustments based on increased monitoring of prothrombin time may be necessary during and after concurrent use.
2. Azothioprine or mercaptopurine:
   Allopurinol-induced inhibition of xanthine oxidase decrease metabolism of these medications and may potentiate therapeutic and toxic effects, especially bone marrow depression. If concurrent use is essential, it is recommended that azothioprine or mercaptopurine dosage be reduced to one fourth of the usual dosage, that the patient be carefully monitored, and that subsequent dosage adjustments be based on patient response and evidence of toxicity. Mercaptopurine may increase serum uric acid concentration in some patients; patients receiving allopurinol (to treat pre-existing hyperuricaemia or gout) may require allopurinol dosage adjustment when mercaptopurine therapy is initiated or discontinued.
3. Urinary acidifiers such as ammonium chloride, ascorbic acid, potassium or sodium phosphate: increased possibility of allopurinol-induced xanthine kidney stone formation.
4. Alcohol, diazoxide, mecalymine or pyrizamine:
   These medications may increase serum uric acid concentrations; dosage adjustment of allopurinol may be necessary.
5. Amoxycillin, ampicillin, bacampicillin or hetacillin:
   Concurrent use may significantly increase the possibility of skin rash.
6. Anti-neoplastic:
   Dosage adjustment of allopurinol may be required during and following of concurrent therapy. Concurrent use of allopurinol with cyclophosphamide and possibly other anti-neoplastic agents may increase the potential for bone marrow depression.
7. Chlorpropamide:
   Allopurinol may inhibit renal tubular secretion of chlorpropamide; patients receiving the medications concurrently should be monitored for possible hyperglycaemic effects.
8. Dacarbazine:
   Additive hypouricaemic effect when used concurrently.
9. Diuretics, thiazide:
   Possibility of severe hypersensitivity reactions.
10. Probenecid, sulphinpyrazone:
    Additive anti-hyperuricaemic effects during concurrent use.
11. Vidaraline, systemic:
    Concurrent use may increase the risk of neurotoxicity, and other side-effects such as anaemia, nausea, pain and pruritus.
12. Xanthines such as aminophylline, theophylline:
    Concurrent use of large doses (600 mg per day) of allopurinol with the xanthines may decrease theophylline clearance, resulting in increased serum concentrations. Theophylline dosage may require adjustment.

Effects on ability to drive and use machines
Since adverse reactions such as somnolence, vertigo and ataxia have been reported in patients receiving allopurinol, patients should exercise caution before driving, using machinery or participating in dangerous activities until they are reasonably certain that allopurinol does not adversely affect performance.

DOSAGE AND ADMINISTRATION
To be taken after food.

attn	customer **Pharmaniaga Manufacturing Berhad**	date **26.05.2022**
1 colours	process colour: Black	**Please Chop & Sign For Approval**
size / description	(L)**297** x (W)**210mm** / **Allopurinol Insert - Front (PRP 0176.12 250522)**	material finishing: **60gm simili**
FocusPrint SDN BHD	t: 603-8766 6030 f: 603-8766 6033 (Factory) website: www.focusprint.com.my	
artwork prepared by: cynthia yap	email: graphic@focusprint.info (graphic Dept)	date:

ARTWORK LOG

Revision no.	Date	Reason for Change
01	22.07.2020	- New artwork received.
02	27.07.2020	- Amendment on wording.
03	27.07.2020	- Rearrange underline and spacing.
04	20.08.2020	- Add spacing in 300mg and fullstop.
05	15.04.2022	- Amend Logo® and wording.
06	26.05.2022	- Amend wording and date of revision.

IMPORTANT ! Please note that this colour print is for visual purpose only. Output colour might differ in actual production.

(L)297mm

Back

(W)210mm

Adult:
Antigout
Initial:
100 mg once daily to be increased by 100 mg per day at one-week intervals until the desired serum uric acid concentration is obtained.

Maintenance:
100 to 200 mg two to three times daily or 300 mg as a single daily dose. The usual maintenance dose is 200 or 300 mg per day in mild gout or 400 to 600 mg per day in moderate severe tophaceous gout.

Neoplastic disease therapy
Initial:
600 to 800 mg per day starting twelve hours to three days (preferably two to three days) prior to initiation of chemotherapy or radiation therapy.

Maintenance:
Dosage should be based on serum uric acid determinations performed approximately forty-eight hours after initiation of allopurinol therapy and periodically thereafter. Allopurinol should be discontinued following the period of tumour regression.

Anti-urolithic (uric acid calculi)
100 to 200 mg daily as a single dose or in divided doses.

Children:
Neoplastic disease
10 to 20 mg per kg body-weight daily.

Renal Failure:
Because oxypurinol is excreted primarily by the kidneys, accumulation may occur in patients with renal failure. Patients receiving dialysis may require therapeutic doses of allopurinol. For patients not receiving dialysis, creatinine clearance may used as a guide but are perhaps limited by the unreliability of creatinine clearances of this low order.

Creatinine clearance (mL/min)	**Dose**
10 to 20	200 mg daily.
3 to 10	Max. of 100 mg daily.
Less than 3	100 mg at intervals of more than 24 hours may be necessary.

Some patients with renal function impairment may require even lower doses of longer intervals between doses. In some cases, 300 mg twice a week or even less, may suffice. Alternatively, it is suggested that if facilities are available for monitoring, the dose should be adjusted to maintain plasma-oxypurinol concentrations below 100 µmmol per litre.

OVERDOSAGE
Symptom: Extended effect of the adverse reactions.

Treatment: For hypersensitivity reactions, glucocorticoids should be administered. Prolonged administration may be required after a severe reaction. For acute poisoning, medication must be discontinued and gastric lavage performed, if large quantities have been ingested. Hydration must be maintained and the patient monitored and observed symptoms treated.

ROUTE OF ADMINISTRATION
Oral.

STORAGE CONDITIONS
Store below 30°C.
Protect from light.

SHELF LIFE
Product should not be used beyond the expiry date imprinted on the product packaging.

PRESENTATION
In blisters of 100 tablets.

Date of revision: 25th May 2022

PRODUCT REGISTRATION HOLDER/MANUFACTURER:
PHARMANIAGA MANUFACTURING BERHAD
(198001006232)
No. 11A, Jalan P/1, Kawasan Perusahaan Bangi,
43650 Bandar Baru Bangi, Selangor Darul Ehsan, Malaysia

PRP 0176.12 250522

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